CropCast: Crop Yield Prediction Using Weather and Pesticide Data

Internship Major Project Report

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Internship role: Advanced AI Development Intern

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Implementation scope: Educational prototype based on the executed outputs and screenshots supplied during development. This report documents the implemented subset, not a full reproduction of the supplied research framework.

Abstract

CropCast is an educational prototype for estimating crop yield and comparing prediction methods. It uses a public dataset containing crop yield, rainfall, pesticide use, temperature, country, and year. The project selects six crops in India and examines the records before modelling. After crop selection and exact-duplicate removal, 2,538 rows remain, but these represent only 138 crop-year groups with one yield outcome per group. The project therefore constructs an annual table with one record per crop-year. Temperature values are averaged within each group as an explicit prototype assumption; this average is not verified as a national temperature measurement.

Linear Regression and Gradient Boosting are compared with crop-average and last-known-yield baselines. Three earlier-year validation periods are used, followed by reporting an exploratory comparison for 2009–2013. The last-known-yield baseline achieves the lowest mean validation MAE, 0.4081 tonnes/ha, and the lowest exploratory-test MAE, 0.5649 tonnes/ha. The tested machine-learning models do not consistently outperform the historical baselines. A Google Colab form displays the historical baseline and ML estimates separately. The prototype demonstrates data preparation, time-based evaluation, and transparent reporting rather than a verified operational forecasting service.

Chapter 1: Introduction

1.1 Project Background

Crop-yield estimation studies the relationship between historical harvest outcomes and available explanatory variables. The supplied base paper investigates rainfall, temperature, pesticide information, and crop type using machine-learning methods [1]. CropCast adapts this topic into a small, reproducible educational project in Google Colab.

1.2 Problem Statement

A complex prediction method is useful only if it performs better than a reasonable simpler method under a meaningful evaluation. Repeated rows and large differences in typical yield between crops can make pooled scores look stronger than the ability to predict changes within one crop. This project examines whether two ML models improve on simple historical baselines while keeping later-year outcomes separate from model fitting.

1.3 Objectives

- Inspect and prepare the supplied dataset for six crops in India.
- Identify exact duplicates and repeated crop-year outcomes.
- Compare Linear Regression and Gradient Boosting with historical baselines.
- Evaluate errors using time-ordered splits and clearly defined units.
- Provide a notebook-based prediction demonstration and report the limitations.

1.4 Scope of the Project

The selected crops are Potatoes, Rice (paddy), Sorghum, Soybeans, Sweet potatoes, and Wheat. The annual table contains 23 observed years between 1990 and 2013; 2003 is absent. The system uses country-level historical records, not farm-specific measurements. It does not provide pesticide recommendations, live weather forecasts, a login system, or a production website.

1.5 Expected Outcomes

The intended outputs are a runnable notebook, a prepared annual dataset, measured comparison tables, a validation chart, a prediction form, and a report. Model superiority is not assumed as an expected outcome; the evaluation determines which approach performs better.

Chapter 2: System Overview

2.1 Existing System

The supplied research paper describes a crop-yield prediction framework using data preparation, regression models, cross-validation, and parameter tuning [1]. It serves as a reference, not as evidence that CropCast reproduces its results. The supplied 10-page companion document discusses an additional voting ensemble but contains inconsistent classifier/regressor terminology. This implementation uses regressors for the numerical yield target and does not implement the voting ensemble or KNN.

2.2 Proposed System

CropCast compares two regressors against two non-ML historical baselines. It first investigates data repetition, then prepares an annual crop-year table. Models use crop type, rainfall, pesticide totals, and the mean of available source temperature values. Year controls the evaluation splits but is not an input to the two fitted regressors.

2.3 System Modules

1. CSV upload and inspection.
2. India and crop selection.
3. Exact-duplicate checks and annual aggregation.
4. Time-based training and validation.
5. Regression and baseline comparison.
6. Result visualization and export.
7. Interactive notebook demonstration.

2.4 Overall Workflow

Upload CSV → inspect columns → filter India and six crops → remove exact duplicates → inspect crop-year groups → construct annual records → form time-ordered splits → train and compare methods → display results and example estimates → export results.

Chapter 3: Literature Survey

3.1 Research Papers Reviewed

Hoque et al. (2024) investigate crop-yield prediction using meteorological and pesticide data, including KNN and Gradient Boosting, with preprocessing and parameter tuning [1]. Their work motivates the subject and the choice of a Gradient Boosting comparison. Their numerical results are not presented as CropCast results.

The supplied companion document, “Incorporating Meteorological Data and Pesticide Information to Forecast Crop Yields Using Machine Learning — Paper.docx.pdf,” describes Linear Regression, KNN, Gradient Boosting, and a voting ensemble. Its author and publication details are not established in the supplied copy, so it is treated as teaching material rather than an independently verified published study.

3.2 Existing Solutions and Comparative Analysis

The base paper explores several ML approaches and hyperparameter optimization. CropCast uses a smaller model set and emphasizes historical baselines, crop-year record inspection, and time-ordered evaluation. This is a difference in implementation scope, not a claim of research novelty or superiority over the paper.

3.3 Research Gap / Project Motivation

The project addresses an evaluation question: do weather-and-pesticide regressors improve on simple crop-history predictions in this prepared dataset? A pooled R^2 alone cannot answer that question. The project adds per-crop checks, baseline comparisons, and an explicit account of repeated annual outcomes. This is a focused review of the supplied paper and companion document, not a systematic literature survey. No claim is made that these practices are absent from the research field.

Chapter 4: Technologies Used

4.1 Programming Language

Python is used for data preparation, training, evaluation, and the demonstration.

4.2 Frameworks and Libraries

- pandas: CSV reading, filtering, grouping, and result tables.

- scikit-learn: ColumnTransformer, OneHotEncoder, Pipeline, LinearRegression, GradientBoostingRegressor, cloning, and evaluation metrics.

- Matplotlib: comparison chart.

- ipywidgets and IPython.display: notebook form and displayed output.

- google.colab.files: uploading data and downloading exports.

- zipfile: packaging result files.

4.3 Development Tools

The prototype runs in Google Colab through a web browser. The notebook is saved as CropCast.ipynb. The student-reported Google Colab environment uses Python 3.13.16, pandas 2.2.3, NumPy 2.1.3, and scikit-learn 1.6.1. No GPU requirement or measured training-time claim is made.

Chapter 5: Architecture and Design

5.1 System Architecture Diagram

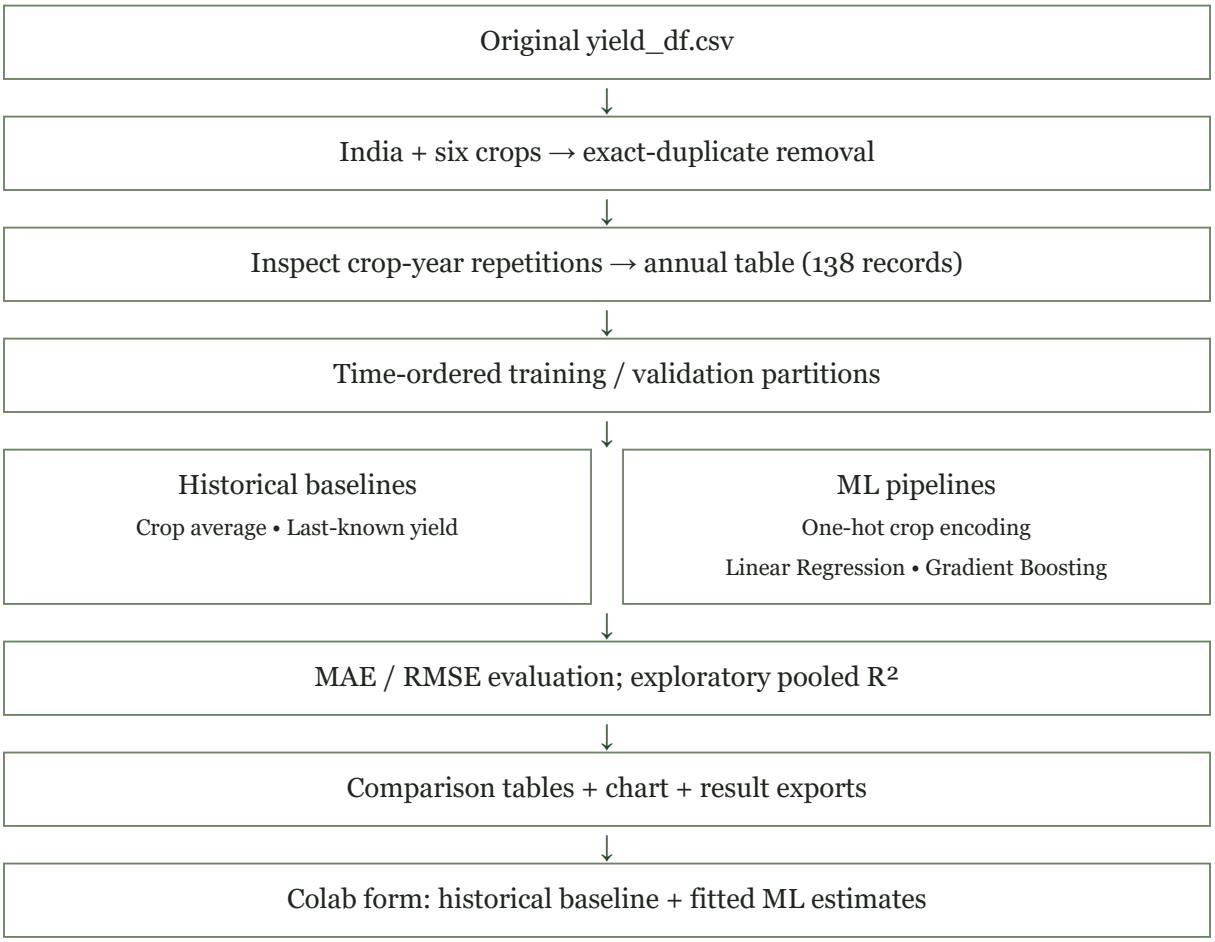


Figure 5.1. Architecture of the implemented educational prototype. Validation uses fresh training-only models; the form uses models fitted before 2009.

5.2 Workflow Explanation

The target is yield in hg/ha. Crop type is one-hot encoded, while numerical predictors are passed through. Encoders and regressors are fitted on the relevant training records only. Each validation round clones and trains a fresh pipeline. Baselines are also computed from that round's training data only.

The form uses the regressors trained before 2009, not models retrained on all available years. Its initial examples come from training records and demonstrate operation rather than unseen-data accuracy. The historical baseline ignores the form's weather inputs.

Chapter 6: Implementation

6.1 Dataset Preparation

The uploaded CSV has 28,242 rows and eight columns, including an obsolete index column, Unnamed: 0. After dropping that column, the India filter gives 4,048 rows and no missing values. Selecting the six project crops gives 3,036 rows. Removing 498 exact duplicates leaves 2,538 rows.

There are 138 distinct crop-year groups. Every group has one yield value, one rainfall value, and one pesticide value, but up to 19 temperature values. The annual table preserves the identical yield, rainfall, and pesticide values and averages temperature values in the cleaned table. This is a descriptive aggregation assumption, not a verified reconstruction of national weather data. The missing 2003 records are not fabricated or interpolated.

6.2 Models and Baselines

Linear Regression is a simple fitted numerical prediction model. Gradient Boosting combines sequentially fitted decision trees. Crop-average baseline predicts the training-period mean for each crop. Last-known-yield baseline predicts the latest available training yield for each crop and holds it constant across each validation/test period.

6.3 Time-Based Evaluation

The annual exploratory split uses 108 training records before 2009 and 30 records from 2009–2013. These later-year results were inspected during development, so they are not described as a pristine final holdout.

Earlier-year validation uses:

Round	Training period	Validation years	Training records	Validation records
1	1990–1998	1999–2001	54	18
2	1990–2001	2002, 2004, 2005	72	18
3	1990–2005	2006–2008	90	18

These are expanding-window retrospective checks. They provide consistency evidence, not an independent confirmation of a design chosen without any later-year inspection.

6.4 Key Features Implemented

The notebook includes dataset checks, annual aggregation, model comparisons, per-crop diagnostics on the preliminary data, validation tables, a chart, and a prediction form. The form checks finite numeric input, rejects negative rainfall/pesticide totals, warns about global training-range violations, and labels negative model outputs as physically invalid. These checks do not guarantee a plausible combination of inputs or farm-level applicability. Results and annual data are exported in a ZIP archive; the notebook is downloaded separately.

6.5 Differences from the Supplied Framework

The companion document also describes KNN, a voting regressor using tree-based components, GridSearchCV tuning, a correlation matrix, geographic exploration, and a web dashboard. These have not been implemented in CropCast. One-hot encoding replaces label encoding for crop categories. Expanding-window validation replaces shuffled K-fold evaluation because years have an ordering. CropCast reports MAE, RMSE, and exploratory R^2, but has not exported an MSE comparison. The interface runs inside Colab rather than as a separate website. Whether this simplified scope satisfies the internship assessment requires mentor confirmation.

Chapter 7: Results and Discussion

7.1 Screenshots

```
Annual dataset rows: 138
Records per crop:
Item
Potatoes          23
Rice, paddy       23
Sorghum           23
Soybeans          23
Sweet potatoes    23
Wheat             23
dtype: int64
```

	Item	Year	yield_hg_ha	rainfall_mm	pesticides_tonnes	mean_source_temperature_c
0	Potatoes	1990	157136	1083.0	75000.0	25.643889
1	Potatoes	1991	162540	1083.0	72133.0	25.811111
2	Potatoes	1992	160294	1083.0	70791.0	25.775789
3	Potatoes	1993	149024	1083.0	66388.0	25.944737
4	Potatoes	1994	166101	1083.0	61357.0	25.764118

Figure 7.1. Prepared annual dataset: 138 records, with 23 observed years for each of six crops.

```
ERRORS FOR EACH VALIDATION ROUND
 Round                     Model  MAE (tonnes/ha)  RMSE (tonnes/ha)
     1     Crop-average baseline           0.5468            0.8848
     1 Last-known-yield baseline           0.7634            1.5051
     1         Linear Regression           0.4876            0.7914
     1         Gradient Boosting           0.7634            1.4793
     2     Crop-average baseline           0.5239            0.9484
     2 Last-known-yield baseline           0.2004            0.3335
     2         Linear Regression           0.5804            0.8820
     2         Gradient Boosting           0.4744            0.9506
     3     Crop-average baseline           0.4096            0.6542
     3 Last-known-yield baseline           0.2604            0.6068
     3         Linear Regression           0.6080            0.7195
     3         Gradient Boosting           0.2868            0.5985

AVERAGE ERRORS ACROSS THREE ROUNDS
                           MAE (tonnes/ha)  RMSE (tonnes/ha)
Model
Last-known-yield baseline           0.4081            0.8151
Crop-average baseline               0.4934            0.8291
Gradient Boosting                   0.5082            1.0094
Linear Regression                   0.5587            0.7976
```

Figure 7.2. Errors for all three validation rounds and their arithmetic averages, in tonnes/ha.

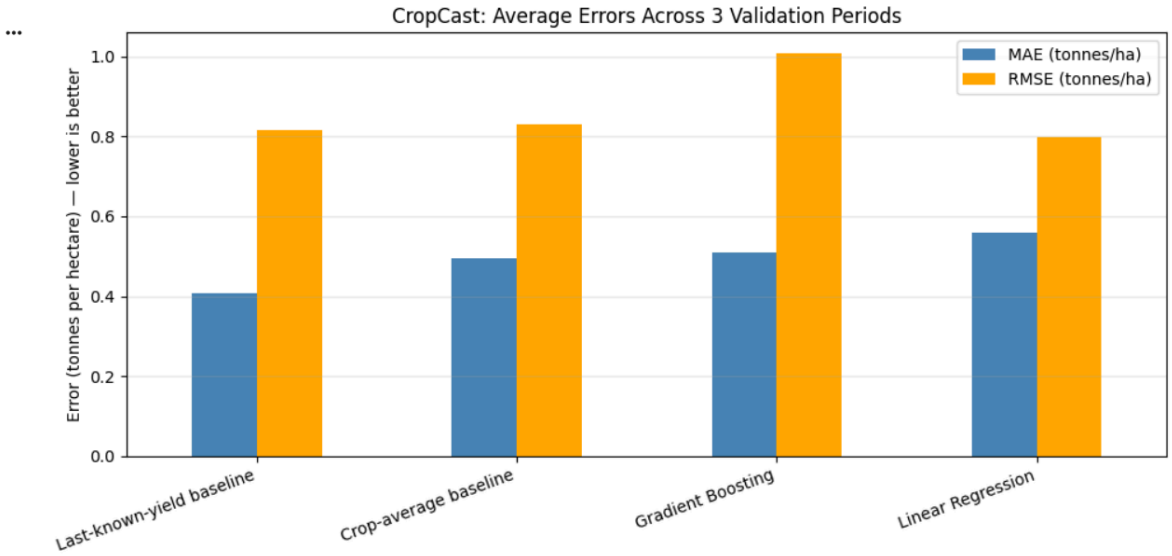


Figure 7.3. Average round-level MAE and RMSE across three earlier-year validation periods. Lower values indicate smaller errors; the last-known-yield baseline has the lowest mean MAE.

Model	MAE (tonnes/ha)	RMSE (tonnes/ha)	R²
Crop-average baseline	1.0037	1.7639	0.9394
Linear Regression	1.0223	1.7917	0.9375
Gradient Boosting	1.0214	1.8676	0.9321
Last-known-yield baseline	0.5649	1.0626	0.9780

Figure 7.4. Annual-data exploratory results for 2009–2013. Last-known yield has the smallest MAE and RMSE in this comparison.

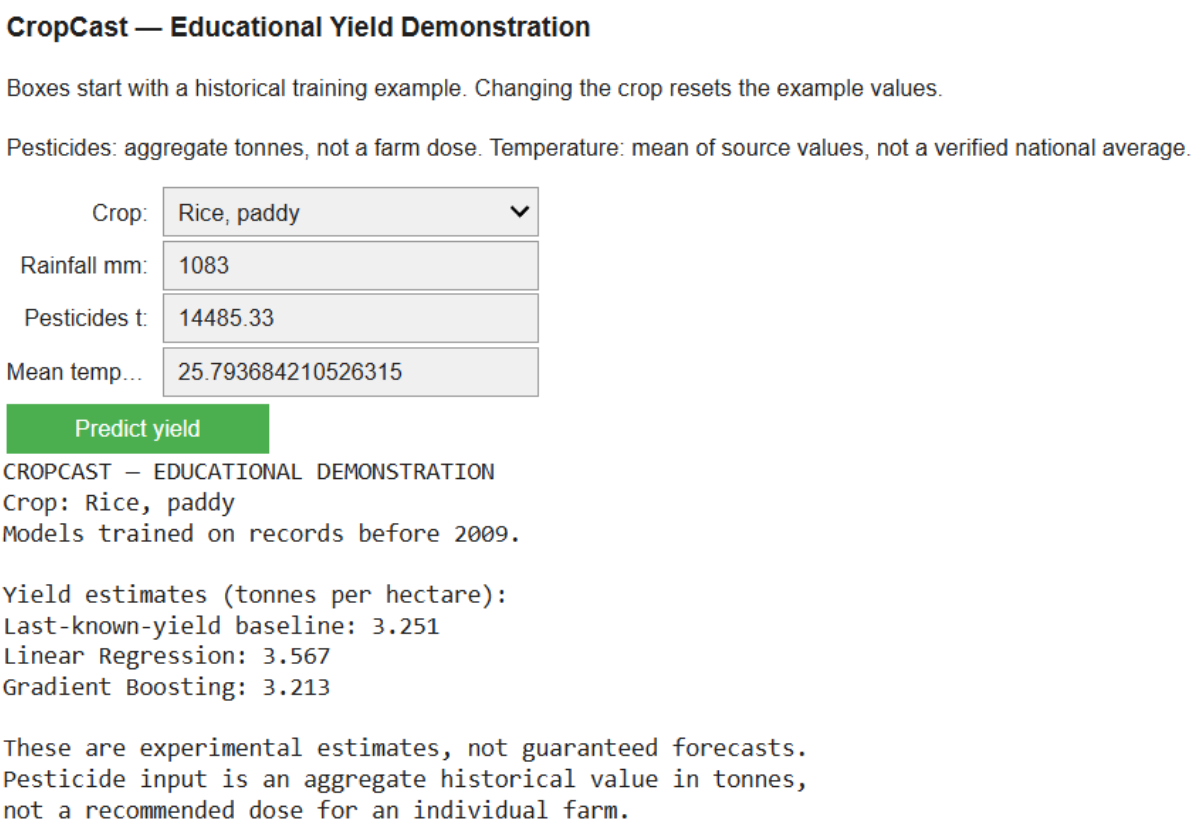


Figure 7.5. Notebook prediction form using a historical rice training example. These outputs demonstrate operation, not unseen-data accuracy.

The figures document the executed annual-data analysis and notebook interface.

7.2 Evaluation Measures

MAE = mean(|actual – predicted|). RMSE = square root of mean((actual – predicted)²). $R^2 = 1 –$ sum((actual – predicted)²) / sum((actual – mean(actual))²). MAE and RMSE are reported in tonnes/ha, using hg/ha ÷ 10,000. Lower MAE/RMSE is better. R^2 is dimensionless, may be negative, and is not percentage accuracy.

7.3 Validation Results

Average of the three round-specific metrics, in tonnes/ha:

Method	Mean MAE	Mean RMSE
Last-known-yield baseline	0.4081	0.8151
Crop-average baseline	0.4934	0.8291
Gradient Boosting	0.5082	1.0094
Linear Regression	0.5587	0.7976

The last-known-yield baseline has the lowest mean MAE. Linear Regression has the lowest mean RMSE. These are arithmetic averages of round-level metrics, not a pooled RMSE computed over all validation predictions. Linear Regression has the lowest MAE in Round 1, while last-known yield has the lowest MAE in Rounds 2 and 3. No method wins every round.

7.4 Exploratory Later-Year Results

Annual-record evaluation on 2009–2013:

Method	MAE (tonnes/ha)	RMSE (tonnes/ha)	Pooled R^2
Crop-average baseline	1.0037	1.7639	0.9394
Linear Regression	1.0223	1.7917	0.9375
Gradient Boosting	1.0214	1.8676	0.9321
Last-known-yield baseline	0.5649	1.0626	0.9780

Last-known yield performs best on all three reported measures in this exploratory split. High pooled R^2 must not be described as percentage accuracy or proof of good within-crop forecasting.

7.5 Output Analysis and Performance Evaluation

The tested ML methods do not consistently beat historical baselines. This finding does not establish that weather is unimportant; the limited dataset, aggregation assumptions, selected features, model settings, and evaluation periods restrict that conclusion. All 138 annual records have rainfall fixed at 1083 mm. Therefore this dataset provides no variation from which to learn a rainfall effect; changing rainfall in the form is outside the observed data support and must not be interpreted as a validated rainfall-response experiment. Since the temperature origins and weighting are unresolved, the annual weather representation remains uncertain. Country-level pesticide totals are not farm dosages or evidence of causal yield effects.

The input form's rice example displays 3.251 tonnes/ha for last-known yield, 3.567 for Linear Regression, and 3.213 for Gradient Boosting. These are estimates for an existing training example, not a new test result. The interface shows method differences without asserting that any estimate is

correct.

7.6 Testing Results and Remaining Checks

Observed checks include successful CSV upload, column inspection, missing-value inspection, duplicate and crop-year checks, time-ordered split assertions, model execution, chart generation, form operation with the rice example, and export download. Full validation of error paths in the form, interactive form error-path testing remains to be performed. The student subsequently confirmed that all cells completed without errors after resetting the Colab runtime and re-uploading the original CSV, and that Predict yield displayed estimates. This fresh-session success is student-reported; the newly executed notebook has not been independently inspected. Software versions and a static review of the downloaded notebook have been recorded. The uploaded later notebook, CropCast (1).ipynb, contains 22 code cells, passes a static Python syntax check, and contains no saved error outputs. Its code includes the reported annual models, validation, chart, form, and export logic. The results ZIP passes an archive integrity check and its metric tables match the report. These checks do not constitute a fresh execution of the notebook. No inference-speed benchmark or production deployment test has been completed.

Chapter 8: Challenges and Learning Outcomes

8.1 Challenges Faced

Repeated crop-year outcomes initially created a misleadingly large impression of sample size. Strong pooled scores also concealed weak crop-specific performance in the preliminary repeated-row evaluation. The supplied reference documents contain inconsistent terminology, and the source temperature values lack verified geographic interpretation in the uploaded table.

8.2 Solutions Implemented

The project introduces explicit crop selection, exact-duplicate removal, crop-year diagnostics, a documented annual aggregation, historical baselines, and time-ordered validation. The report separates measured outputs from claims made in reference papers. The form labels aggregate inputs and experimental estimates.

8.3 Technical Skills Learned

The work practises pandas filtering and grouping, categorical encoding, pipelines, train/validation separation, regression metrics, baseline comparisons, plotting, notebook widgets, and result export.

Development used AI-guided explanations and code assistance. The student should verify the code, understand the methods, and follow the institution’s disclosure policy.

Chapter 9: Future Enhancements

9.1 Proposed Improvements

Verify the origin and meaning of each source record, particularly temperature and country-level pesticide totals. Obtain a documented annual or geographically matched dataset and preserve its provenance. Evaluate genuinely new years or another appropriate independent dataset before making forecasting claims.

9.2 Additional Features

Potential extensions include per-crop error plots, a documented recent-history/trend model, uncertainty intervals with a suitable validation method, stronger input checks, and a downloadable prediction record. These are proposals, not implemented features.

9.3 Scalability Opportunities

A future implementation could use larger verified datasets and geographically defined models. Packaging as an independent application is possible but has not been implemented. More algorithms alone do not solve poor data quality or inadequate evaluation.

Chapter 10: Conclusion

10.1 Summary of Work Completed

CropCast prepares a six-crop India dataset, builds an annual crop-year table, compares two ML regressors with two historical baselines, and presents a notebook demonstration with exportable results.

10.2 Achievements

The project identifies the difference between repeated source rows and distinct annual yield outcomes. It documents why pooled scores need careful interpretation and compares predictions against simple methods instead of assuming model complexity guarantees improvement.

10.3 Project Impact

Its demonstrated contribution is educational and methodological. The measured results favour last-known yield by mean validation MAE and by the reported exploratory-test measures. The project does not establish farm-level reliability or superiority of weather-and-pesticide ML forecasts.

10.4 Final Remark

The main lesson is that honest evaluation and clear data assumptions are essential parts of a prediction project. A simple baseline can outperform a more complex model, and that result should be reported rather than hidden.

References

[1] M. J. Hoque et al., “Incorporating Meteorological Data and Pesticide Information to Forecast Crop Yields Using Machine Learning,” IEEE Access, vol. 12, 2024. DOI: 10.1109/ACCESS.2024.3383309. Source: supplied 19-page paper.

[2] Rishi Patel, “Crop Yield Prediction Dataset,” Kaggle. https://www.kaggle.com/datasets/patelris/crop-yield-prediction-dataset . Dataset page checked during this project on 23 October 2026. The page attributes underlying sources to FAO and World Bank. Original source-record joins and temperature weighting have not been independently verified.

[3] Supplied course companion document, “Incorporating Meteorological Data and Pesticide Information to Forecast Crop Yields Using Machine Learning — Paper.docx.pdf,” 10 pages. Author and publication details not established in the supplied copy.

Project results source: student-shared Google Colab outputs in this conversation on 23 October 2026. Final figures should be checked against the submitted notebook and exported result files.

Numerical Reproducibility Check

A separate numerical rerun using the original CSV reproduced the 138-record annual table, all exploratory later-year scores to four decimals, the rice example estimates, and baseline/Linear Regression validation scores. Gradient Boosting validation differed slightly: mean MAE 0.5062 and mean RMSE 1.0043 locally, compared with 0.5082 and 1.0094 in the saved Colab outputs. The difference arose in validation Round 1; its cause is not established. The local environment was Python 3.12.7, pandas 3.0.5, NumPy 2.5.1, and scikit-learn 1.9.1; the student-reported Colab versions are Python 3.13.16, pandas 2.2.3, NumPy 2.1.3, and scikit-learn 1.6.1. Environment differences are a possible explanation for the numerical difference, but have not been established as its cause. The tables and screenshots above retain the original Colab results. This rerun excluded Colab upload, plotting, widget interaction, and browser downloads, and does not certify the complete Colab interface.